

COMPUTER Networks

Assignment COPY

Assignment COPY

Name = Suman Kushwaha

Subject = Computer Networks

Semester = 4th

GitHub = Kushwaha-Suman

Website = KushwahaSuman.Com.NP

1) Explain the seven layers of the OSI model in detail with suitable diagrams.

→ The OSI (Open Systems Interconnection model) model is a conceptual framework that explains how data travels from one computer to another through a network. It divides the communication process into 7 layers, where each layer performs specific functions.

Easy way to remember the layers.

Top to Bottom (7 → 1):

All people seem to need data processing

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Datalink
- 1 Physical

1) Physical Layer:

→ The first layer of OSI model is called Physical layer. The physical layer describes electrical or optical signals. All the physical connectivity of network takes place such as cable & RJ45, clamping. It converts data into bits and forwards it to the physical layer to the receiving side.

Devices used:

- HUB
- Repeater

2) Data-link Layer:

→ The second layer of the OSI model is called data-link layer. The physical to data link layer establishes a connection between two physically connected devices on a network.

→ It is responsible for node-to-node delivery of data. The data link layer consists of two sub-layers.

LLC (logical link control) sublayer

MAC (media access control) sublayer.

LLC :

→ It is responsible for transferring the packets to the network layer of the receiver that is receiving.

MAC

→ Establishes and maintains link between communicating devices

Protocol used under this layer are: PPP (Point to Point Protocol), PPTP (Point to Point Tunneling Protocol), ARP (Address Resolution Protocol) and RARP (Reverse address resolution protocol).

Devices: Switch.

3) Network layer:

→ The 3rd layer of OSI network model is called network layer. The network layer is responsible for the source to destination delivery of a packet.

1. logical addressing:

→ logical address provides a unique address that identifies computer on internet.

2 Routing

→ Router determines the best path to send and receive data from source to destination. Protocol used under the layer are: IP, RIP, IGRP, EIGRP, OSPF etc.

Devices,

- Router

4) Transport layer

→ the 4th layer of OSI model is called transport layer. It is responsible for process to process delivery of the entire message.

1 TCP/IP: it established the connection between hosts

2 UDP: it is unreliable protocol.

The Transport layer has following functionality

1. Segmentation
2. Flow control
3. Error control
4. Sequencing

5) Session layer:

→ The 5th layer of OSI model is called session layer. The session layer is responsible for dialog control, establishing the connection, manage the connection and terminate the connection. Sessions layer also controls whether to send the data in simplex, half duplex and full duplex.

- Full Duplex

→ Simultaneous two-way communication

- Half-Duplex:

→ two-way communication, but not simultaneous

- Simplex

→ one way communication.

6) presentation layer:

→ The 6th layer of OSI model is called presentation layer. This layer is responsible for translation, encryption and compression. The protocol works under this layer is SSL.

7) Application Layer:

→ This is the final layer of OSI model. This layer provides user interface and support services such as mail services, file transfer, browsing internet, remote desktop connection etc. The protocols that communicate with the application layer are SMTP, HTTP, POP3, IMAP, DHCP, & FTP.

2) Explain the TCP/IP model and describe all its layers in details.

→ TCP/IP model consists of four layers:

4 Application layer

3 Transport layer

2 Internet layer

1 Network Access layer

1) Network Access Layer

→ This layer is the lowest layer of TCP/IP model.

→ The network access layer describes how data is sent across the network.

→ This layer is a combination of the data link layer and the physical layer defined in the OSI model.

→ This layer is responsible for the transmission of data between two devices in the same network.

→ The function of this layer is to encapsulate the IP datagram transmitted by the network into frames and map the IP address to the physical address.

→ The protocols used by this layer are: Ethernet, FDDI (Fiber Distributed Data Interface), X.25

2) Internet layer:

→ The data is packaged in the form of IP datagram, this datagram contains the Source and destination IP address so that the data can be easily sent & received.

→ It is also called network layer.

The following protocols are used by this layer:-

IP protocol: It's full name is Internet protocol and its main function is to deliver packets from source to destination. It has two versions IPv4 & IPv6.

ARP:

→ It's full name is Address Resolution Protocol. It's task is to find the physical address from the IP address.

ICMP:

→ It's full name is Internet Control Message Protocol. It's task is to inform the host about the problems coming in the network.

3) Transport Layer:

→ This layer is responsible for the transmission of data. It is also responsible for the reliability, flow control and correction of the data.

Two main protocols work in this layer:

- 1 Transmission Control Protocol (TCP)
- 2 User Datagram Protocol (UDP)

4) Application layer:-

→ This layer is concerned with providing network service to the applications.

→ This layer provides communication to the user for example; through web browser, e-mail, and other applications.

→ The application layer sends data to the transport layer and receives data from it.

→ This layer allows the user to interact with the application.

The following are the protocols used in the application layer:

HTTP & HTTPS:

→ It is used to access data on internet and it is also used to transfer text, audio & video.

SNMP:

- Its full name is Simple network management protocol.
- It is a framework used to manage devices on the Internet.

SMTP:

- Its full name is Simple mail transfer protocol. It is used to send data from one e-mail, to another e-mail address.

DNS

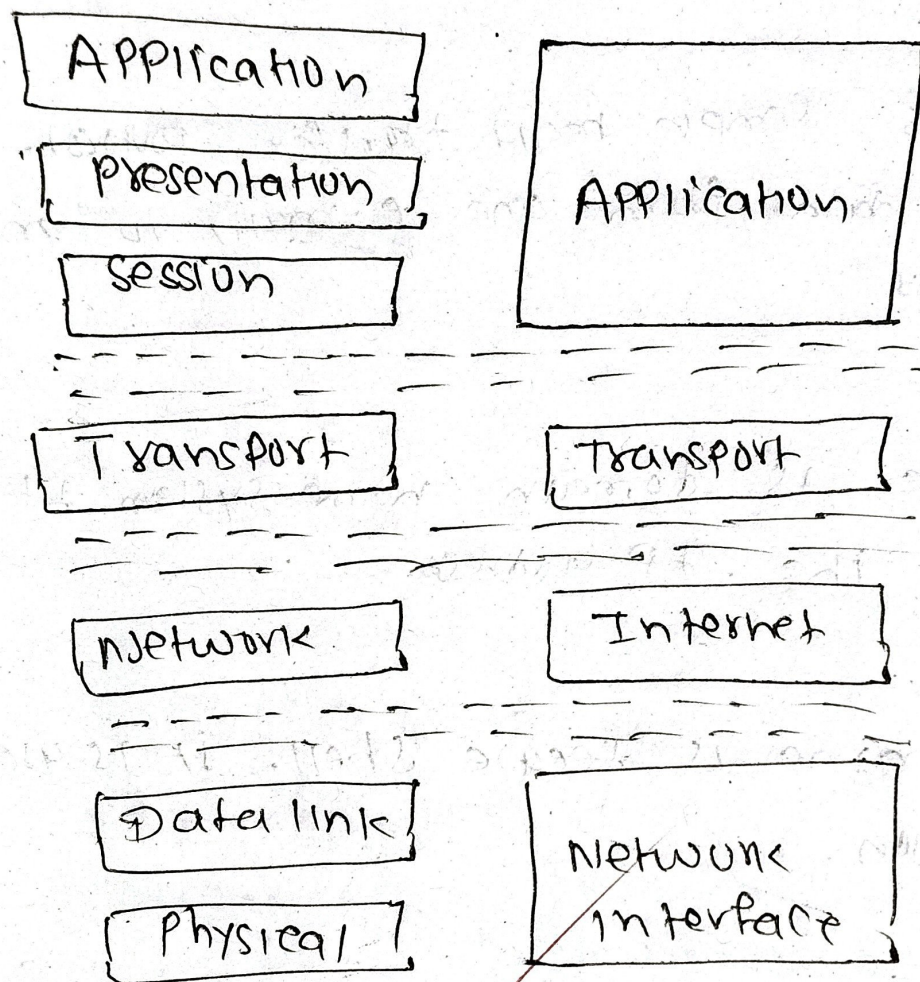
- Its full name is domain name system. It is used to map the IP address.

SSH:

- Its full name is Secure Shell. It is used for encryption.

3) compare the OSI model and TCP/IP model with a suitable diagram.

→ OSI vs TCP/IP



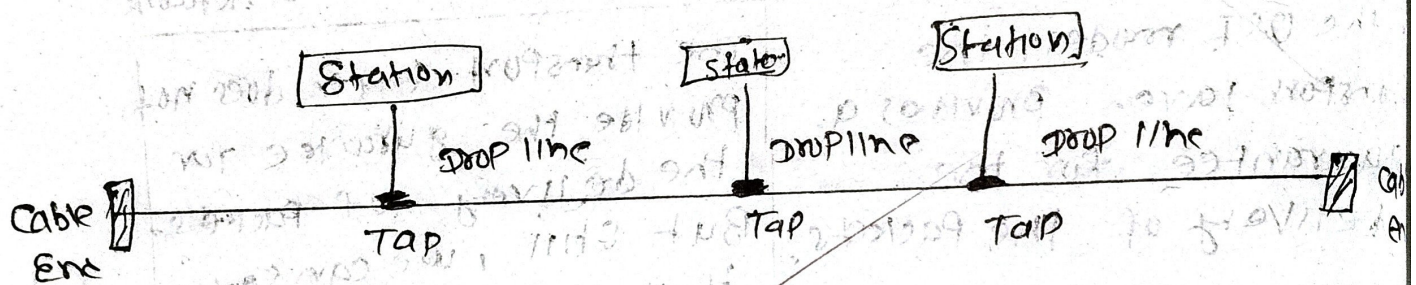
OSI Layer	TCP/IP Layer
The full form of OSI is Open Systems Interconnection.	The full form of TCP/IP is Transmission Control Protocol Internet Protocol.
OSI model has been developed by ISO (International Standard Organization).	It was developed by ARPANET (Advance Research project Agency network).
The OSI model is a conceptual model that characterizes & standardizes the communication function of computing system.	TCP/IP is most widely used communication protocols, TCP/IP prepares and forwards data packets over a network.
In the OSI model, the transport layer provides a guarantee for the delivery of the packets.	The transport layer does not provide the guarantee for the delivery of packets. But still, we can say that it is a reliable model.
This model is based on vertical approach.	This model is based on a horizontal approach.
Protocols in the OSI model are hidden and can be easily replaced when the technology changes.	In this model, the protocol cannot be easily replaced.
Network layer is both connection oriented & connection less.	Network layer is connection less.
It is protocol dependent.	It is protocol independent.
It has 7 layers.	It has 4 layers.

4) Explain different types of network topology with diagrams and advantage / disadvantages.

→ The arrangement of physical connectivity entire network is called topology. The different topologies are given below.

1) Bus topology:

→ A Bus topology in which all computer are connected in a single cable or a single line is called bus topology or it is also called linear topology.



Advantages:

- It is easy to set-up and extend bus network.
- Cable length required for this topology is the ~~best~~ least compared to other network.
- Bus topology very cheap.
- Linear Bus network is mostly used in small networks.

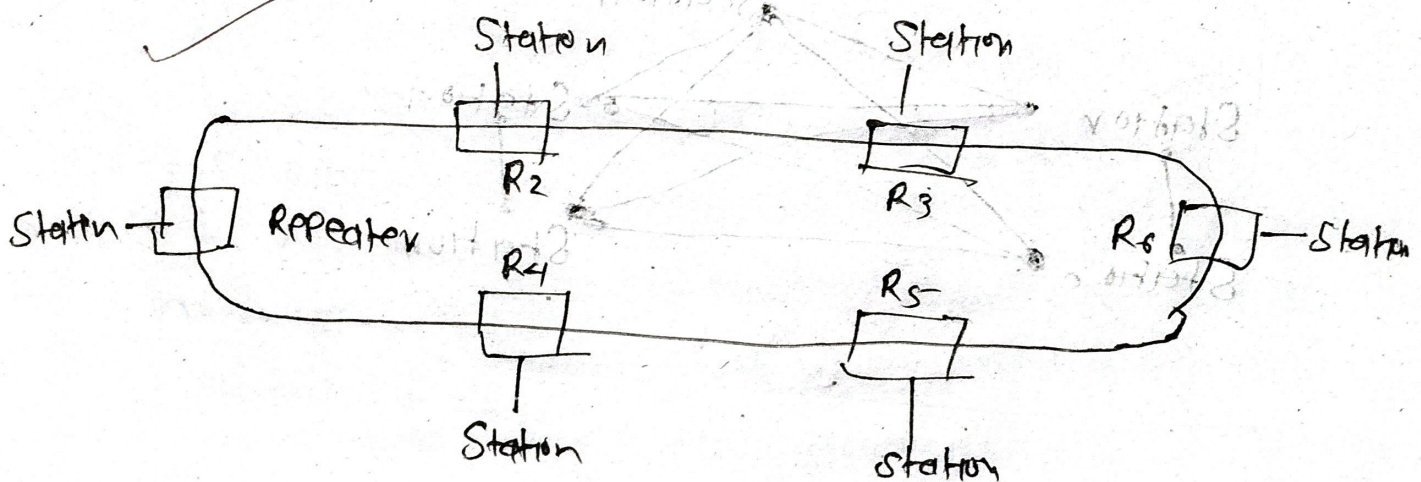
Disadvantage :

- It can be hard to troubleshoot individual device
- Bus topology is not great for large network.
- Terminators are required for both ends of the main cable
- If a main cable is damaged, the network fails or splits into two.

2) Ring Topology :

- A topology in which all computers are connected in a circular ring is called ring topology.

There are no terminated ends. A ring topology connects one host to the next and the last host to the first. The signal travels around the loop in one direction and pass through each computer.



Advantages:

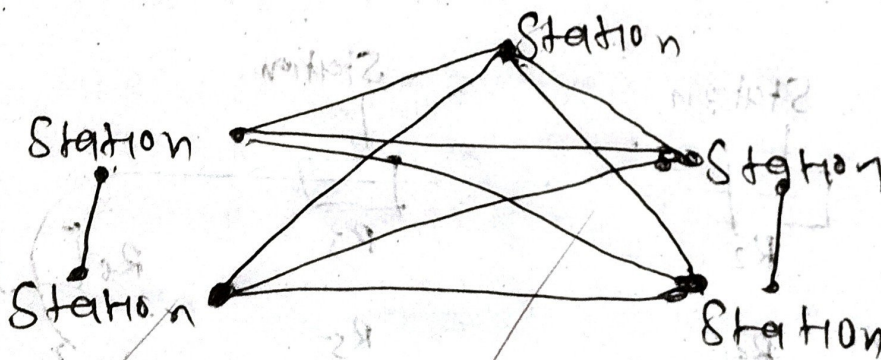
- All data flows in one direction, reducing the chance of packet collision
- A network server is not needed to control network connectivity between each workstation.
- Data can transfer between workstations at high speeds.

Disadvantages:

- Each packet of data must pass through all the computers between source and destination. This makes it slower than star topology
- If one workstation or port goes down, the entire network gets affected.

3) mesh topology:

→ A topology in which all computers are connected to each other



Advantages:

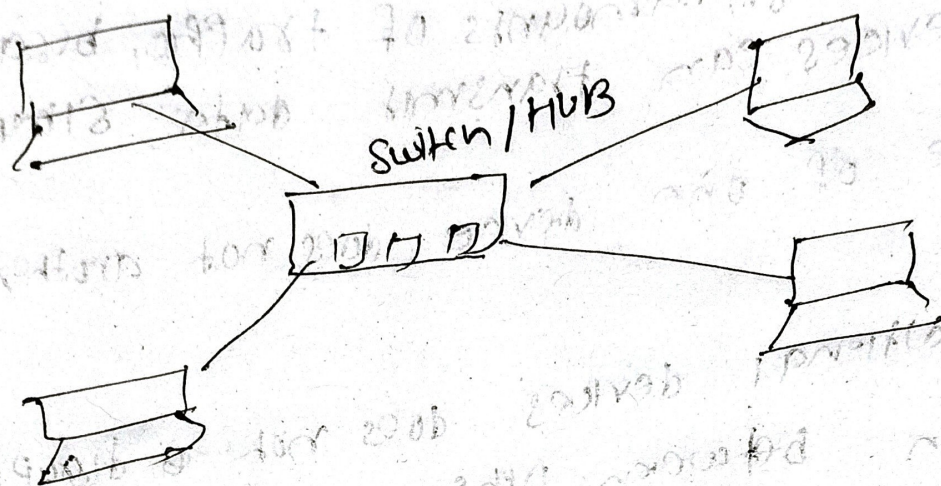
- Can handle high amounts of traffic, because multiple devices can transmit data simultaneously
- A failure of one device does not affect another device
- Adding additional devices does not disrupt data transmission between other devices

Disadvantages:

- It is very expensive
- Building and maintaining the topology is difficult and time consuming

4) Star topology:

- A topology in which all computers are connected centrally by using switch. Signals are transmitted from the sending computer through the hub or switch to all computers on the network. This topology originated in the early days of computing and it is very popular. It is now a common topology in microcomputer networking.



Advantages:

- Centralized management of the network, through the use of the central computer, hub or switch
- Easy to add another computer to the network
- If one computer on the network fails, the rest of the network continues to function normally

Disadvantage:

- Uses more cable
- The use of hub, a router or a switch as central device increase the overall cost of the network
- If the central computer, hub or switch fails, the entire network goes down and all computer are disconnected from the network.

P. Mulla
2025-02-26

Completed 3/7/2025